



AI Explorer Student Workbook

A Personalized Learning Journey Through Artificial Intelligence

“Just like training your parakeets takes patience and observation, learning AI takes curiosity and practice. Just like a great piano performance requires understanding the notes before playing them, AI mastery requires understanding the concepts before applying them. And just like horseback riding demands trust and communication, working with AI demands clear, precise instructions.”

Student Name: _____ **Date Started:** _____

Teacher: _____ **Goal Completion Date:** _____

How to Use This Workbook

This workbook is your companion to the **AI Explorer** online portal. After completing each lesson online, come back here to reinforce what you learned. Each module section contains:

- A **Lesson Summary** to review key ideas
 - **Key Terms** to define in your own words
 - **Fill-in-the-Blank** exercises to test recall
 - **Reflection Questions** to think critically
 - **My Notes** pages for your own thoughts
 - A **Practice Quiz** to prepare for the online checkpoint
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MODULE 1: Introduction to AI

Lesson Summary

Artificial Intelligence (AI) is a branch of computer science focused on building systems that can perform tasks that normally require human intelligence — things like recognising faces,

understanding speech, making decisions, and learning from experience.

Unlike traditional programs that follow fixed rules (like a recipe), AI systems **learn from data**. Think of how you trained your African ring-necked parakeets: you didn't hand them a rulebook — you rewarded the right behaviour and they learned from experience. AI works the same way.

AI is already everywhere: your phone's face unlock, Spotify's song recommendations, Shazam recognising a piano piece you're playing — all AI.

Key Terms — Define in Your Own Words

Artificial Intelligence:

Machine Learning:

Algorithm:

Training Data:

Rule-Based System:

Fill in the Blanks

1. AI stands for _____.
2. Unlike traditional programs, AI systems learn from _____ rather than fixed rules.
3. When you trained your parakeets by rewarding good behaviour, that is similar to how AI uses _____ signals to improve.
4. Machine Learning is a _____ of AI where systems learn patterns from data.
5. Shazam can identify a song you play on the piano because it uses _____ to match audio patterns.

6. A rule-based system follows _____ instructions, while an AI system learns from _____.

Reflection Questions

1. Name three AI tools you use in your daily life. How does each one “learn” or improve?

2. How is training a parakeet similar to training an AI model? What are the differences?

3. Do you think AI could ever learn to compose music the way a pianist does? Why or why not?

My Notes

Module 1 Practice Quiz

Circle the correct answer:

1. What does AI stand for?

- a) Automated Instructions
- b) **Artificial Intelligence**
- c) Advanced Internet
- d) Algorithmic Input

2. Which of the following is an example of AI in everyday life?

- a) A light switch
- b) A printed book

- c) **Spotify recommending songs**
- d) A calculator

3. How is Machine Learning different from rule-based programming?

- a) Machine Learning uses more code
- b) **Machine Learning learns from data instead of following fixed rules**
- c) Rule-based systems are faster
- d) They are the same thing

4. When your phone unlocks using your face, what type of AI is being used?

- a) Rule-based system
- b) **Computer vision / facial recognition AI**
- c) Text generation
- d) None of the above

5. Training a parakeet by rewarding correct behaviour is most similar to:

- a) Writing a computer program
- b) Searching the internet
- c) **Reinforcement learning in AI**
- d) Using a spreadsheet

Score: _____ / 5

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MODULE 2: How AI Actually Works

Lesson Summary

Inside most modern AI systems are **neural networks** — layers of mathematical “neurons” inspired by the human brain. Data flows through these layers, and the network adjusts its internal settings (called **weights**) until it gets good at its task.

Think of it like learning a piano piece: at first you make mistakes, but each time you practice, your brain adjusts. Neural networks do the same thing — they practice millions of times on training data.

Data bias is a critical concept: if the data used to train an AI is unfair or incomplete, the AI will make unfair decisions. For example, if a horse-breed recognition AI was only trained on photos of one breed, it would fail to identify others.

Key Terms — Define in Your Own Words

Neural Network:

Weights (in AI):

Training:

Data Bias:

Overfitting:

Fill in the Blanks

1. Neural networks are inspired by the structure of the human _____.
 2. The internal settings that a neural network adjusts during training are called _____.
 3. If an AI is trained on unfair or incomplete data, it will produce _____ results.
 4. Learning a piano piece through repeated practice is similar to a neural network _____ on training data.
 5. When an AI memorises training data too closely and fails on new data, this is called _____.
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Reflection Questions

1. Imagine you are training an AI to recognise African ring-necked parakeets. What data would you need? What could go wrong if your data was biased?

2. How is learning a new piano piece similar to how a neural network learns? What happens when you “overfit” (memorise) a piece without truly understanding the music?

My Notes

Module 2 Practice Quiz

Circle the correct answer:

1. What are neural networks inspired by?

- a) Computer circuits
- b) **The human brain**
- c) The internet
- d) Spreadsheets

2. What does a neural network adjust during training?

- a) Its programming language
- b) The amount of data
- c) **Its weights**
- d) Its screen resolution

3. Data bias in AI means:

- a) The AI is very fast
- b) **The training data is unfair or incomplete, causing unfair outputs**

- c) The AI uses too much memory
- d) The AI is too accurate

4. Overfitting occurs when:

- a) The AI has too little data
- b) **The AI memorises training data too closely and fails on new examples**
- c) The neural network has too many layers
- d) The AI runs too slowly

Score: _____ / 4

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✨ MODULE 3: Generative AI & LLMs

Lesson Summary

Generative AI creates new content — text, images, music, code, and video — rather than just analysing existing content. The most powerful generative AI tools today are **Large Language Models (LLMs)** like ChatGPT (OpenAI), Claude (Anthropic), and Gemini (Google).

LLMs are trained on enormous amounts of text from the internet, books, and other sources. They learn to predict the next word in a sequence — and by doing this billions of times, they develop an impressive ability to write, reason, and converse.

Prompt engineering is the skill of writing clear, detailed instructions to get the best output from an AI. Just like giving a precise instruction to your horse trainer gets better results than a vague one, a precise prompt gets better AI output.

Key Terms — Define in Your Own Words

Generative AI:

Large Language Model (LLM):

Prompt:

Token:

Hallucination (AI):

Fill in the Blanks

1. ChatGPT, Claude, and Gemini are examples of _____
_____.
2. LLMs generate text by predicting the next _____ in a sequence.
3. Writing clear, detailed instructions to get better AI outputs is called
_____.
4. When an AI confidently states something false, this is called an AI
_____.
5. Generative AI can create _____, _____,
_____, *and* _____.

Prompt Engineering Practice

Improve these weak prompts. Rewrite each one to be more specific and detailed:

Weak: "Write something about horses."

Better:

Weak: "Make a song."

Better:

Weak: "Tell me about parakeets."

Better:

My Notes

Module 3 Practice Quiz

1. What does LLM stand for?

- a) Large Learning Machine
- b) **Large Language Model**
- c) Logical Learning Method
- d) Language Learning Module

2. How do LLMs generate text?

- a) By searching the internet in real time
- b) By copying from books
- c) **By predicting the next word/token based on training**
- d) By following fixed grammar rules

3. An AI “hallucination” means:

- a) The AI creates images
- b) The AI runs slowly
- c) **The AI confidently states something that is false**
- d) The AI asks for more information

4. Which of the following is the BEST prompt?

- a) “Write about AI”
- b) “Tell me stuff”

- c) “You are a science teacher. Explain how neural networks work to a 14-year-old using a piano analogy. Keep it under 200 words.”
- d) “AI explanation please”

Score: _____ / 4

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MODULE 4: Ethics & the Future of AI

Lesson Summary

As AI becomes more powerful, we must think carefully about how it is used. **AI ethics** covers questions like: Is this AI fair? Could it be used to harm people? Who is responsible when AI makes a mistake?

Deepfakes are AI-generated videos or images that make it look like someone said or did something they never did. They can be used to spread misinformation and are a serious ethical concern.

AI is also changing the **job market** — some jobs are being automated, but new jobs are being created too. The key is to develop skills that work *with* AI rather than compete against it.

Key Terms — Define in Your Own Words

AI Ethics:

Deepfake:

Bias (in AI decisions):

Automation:

Reflection Questions

1. If an AI was trained to identify horses for a riding competition but it was biased against certain breeds, what ethical problems could arise?

2. How could deepfakes be used to harm someone? What should you do if you see a suspicious video online?

3. Name two jobs that AI might automate and two new jobs that AI is creating.

Jobs AI Might Automate	New Jobs AI Is Creating
1.	1.
2.	2.

My Notes

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MODULE 5: Connectors, APIs & Integrations

Lesson Summary

APIs (Application Programming Interfaces) are the “bridges” that let different software applications talk to each other. When you use an app that shows weather data, it’s using a weather API. When an AI tool connects to your calendar, it uses an API.

Connectors in AI platforms (like Manus) are pre-built integrations that connect AI agents to external services — email, Notion, Google Drive, Slack, and more — without needing to write

complex code.

Think of APIs like a horse's reins: they give you a standardised way to communicate with and control a powerful system. The right connector lets you guide AI to do exactly what you need.

Key Terms — Define in Your Own Words

API:

Connector:

Integration:

Webhook:

Fill in the Blanks

1. API stands for _____.
2. APIs act as _____ between different software applications.
3. A _____ is a pre-built integration that connects an AI agent to an external service.
4. When two apps share data automatically without manual input, this is called an _____.

My Notes

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MODULE 6: Building with AI — Websites & UI

Lesson Summary

You no longer need to be a professional programmer to build websites. AI tools like **Manus**, **Bolt.new**, **v0 by Vercel**, and **Same.new** can generate entire websites from a text description. You describe what you want, and the AI builds it.

UI (User Interface) refers to the visual elements a user interacts with — buttons, menus, forms, and layouts. AI can now generate professional UI designs from simple descriptions or even sketches.

Same.new is a tool that can clone an existing website's design and let you customise it — like having a blueprint of a building and redesigning the interior.

Key Terms — Define in Your Own Words

UI (User Interface):

UX (User Experience):

Website Cloning:

Responsive Design:

Activity: Design Your Dream Website

Imagine you are building a website for your parakeet breeding hobby. Describe what it would look like:

Website Name:

Purpose:

Main Pages:

Colour Scheme:

Key Features:

Write a prompt you would give to an AI to build this website:

My Notes

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MODULE 7: AI Agents in Action

Lesson Summary

An **AI Agent** is an AI system that can take actions on your behalf — browsing the web, writing emails, running code, managing files, and more. Unlike a simple chatbot that just answers questions, an agent can complete multi-step tasks autonomously.

NotebookLM (by Google) is an AI tool that reads your documents and lets you ask questions about them, create study guides, and generate audio summaries. It's like having a study partner who has read everything.

Gamma.ai creates beautiful presentations and documents from a simple text prompt. Instead of spending hours on PowerPoint, you describe your topic and Gamma builds the slides.

Browser automation lets AI agents control a web browser — clicking buttons, filling forms, and navigating pages — just as if a human were doing it.

Key Terms — Define in Your Own Words

AI Agent:

NotebookLM:

Browser Automation:

Autonomous:

Activity: Plan an AI Agent Task

Describe a task you wish an AI agent could do for you automatically:

Task:

Steps the agent would need to take:

1.

2.

3.

4.

Which tools/connectors would it need?

My Notes

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MODULE 8: Notion & Your AI Productivity Stack

Lesson Summary

Notion is an all-in-one workspace for notes, databases, project management, and wikis. With Notion AI built in, you can ask it to summarise pages, generate content, translate text, and organise information automatically.

A **productivity stack** is the combination of tools you use together to manage your work and life. A powerful AI productivity stack might include: Notion (notes & projects), Gmail with AI (email), Google Calendar (scheduling), and an AI assistant like Claude or ChatGPT.

Other powerful platforms include **Monday.com** (project management with visual boards), **Trello** (kanban-style task cards), **Asana** (team task management), **ClickUp** (all-in-one work hub), and **Airtable** (database-spreadsheet hybrid).

Platform Comparison Table

Fill in what you know about each platform:

Platform	Best For	Key Feature	AI Capability
Notion			
Monday.com			
Trello			
Asana			
ClickUp			
Airtable			

Activity: Build Your Personal Productivity Stack

Which tools would you use for each area of your life? Fill in your ideal stack:

Area	Tool I Would Use	Why
School notes & studying		
Managing horse riding schedule		
Tracking piano practice		
Organising parakeet care		
Personal projects		
Communication		

My Notes

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🌟 BONUS: AI Tools Landscape — The Best Tools for Every Category

The Best AI Tools Right Now

Category	Top Tools	Best For
Chatbots / LLMs	ChatGPT, Claude, Gemini, Copilot	Writing, Q&A, coding help, research
Deep Research	Perplexity AI, NotebookLM, Elicit, Consensus	Academic research, fact-checking
Image Generation	Midjourney, DALL·E 3, Stable Diffusion, Adobe Firefly	Art, design, illustrations
Video Generation	Runway, Pika Labs, Sora, Luma AI, Synthesia	Short videos, animations, avatars
Music Generation	Suno, Udio, Mubert, AIVA	Songs, soundtracks, background music
Coding	GitHub Copilot, Cursor, Replit AI, Codeium	Writing and debugging code
Presentations	Gamma.ai, Beautiful.ai, Tome	Slides, decks, visual reports
Productivity	Notion AI, Monday.com, ClickUp AI	Notes, projects, task management
Website Building	Manus, Bolt.new, v0, Same.new	Building websites with AI
E-commerce	Shopify, PagePilot.ai, Dropified	Online stores, product pages
Email & Communication	Gmail AI, Superhuman, Lavender	Smart email drafting and management
Voice & Speech	ElevenLabs, Whisper, Descript	Voice cloning, transcription, podcasts

Activity: Tool Explorer

Pick **three** tools from the table above that you want to try. For each one, write:

Tool 1:

What I want to use it for:

Tool 2:

What I want to use it for:

Tool 3:

What I want to use it for:

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OVERALL REFLECTION & GOALS

What I've Learned

Write 3–5 sentences summarising the most important things you have learned so far in this course:

My Favourite AI Tool So Far

Tool:

Why I like it:

How I Plan to Use AI in My Life

For school:

For my parakeets / animals:

For piano / music:

For horse riding:

For future career goals:

Questions I Still Have

1.
2.
3.
-

My AI Learning Goals

Goal	Target Date	Completed?
Complete all 8 modules online		☐
Score 80%+ on every quiz		☐
Build my first AI-powered project		☐
Try 5 different AI tools		☐
Teach someone else one AI concept		☐

“The best way to predict the future is to create it — and with AI, you have the tools to create anything.”

AI Explorer Workbook | Prepared for personal educational use | All modules correspond to the AI Explorer online portal.